ACS OMEGA

Open Access

This article is licensed under CC-BY-NC-ND 4.0

http://pubs.acs.org/journal/acsodf

Article

Digital-Twin Enabled "Living Fresco"

Shu Wu, Qi Chen, Chao Hu, Yingjian An, and Feng Han*

Cite This: *ACS Omega* 2026, 11, 8011−8023

Read Online

ACCESS | Metrics & More | Article Recommendations

ABSTRACT: Translucent polymer-concrete claddings are emerging as multifunctional building skins that merge structural performance, daylighting, and human−architecture interaction. However, the lack of integrated design frameworks coupling durability, real-time aesthetics, and long-term service-life prognosis limits their adoption. Here, we present a glass-fiber-reinforced polymer (GFRP) mesh-enhanced translucent polymer-concrete (GFRP-TRPC) panel that simultaneously serves as load-bearing façade, AI-driven light canvas, and self-sensing structural element. Epoxy-vinyl ester polymer concrete with 30 vol % recycled glass aggregate and embedded PMMA optical fibers (ϕ 1 mm) was fabricated by a one-step vacuum-assisted resin infusion process, yielding 30-mm-thin elements with 65 MPa compressive strength and 22% optical transmittance. Bidirectional E-glass GFRP grids (44% fiber volume) restrained microcracking and enabled distributed fiber Bragg-grating (FBG) sensing at 5 cm pitch. A vision transformer−generative adversarial network (ViT-GAN) converted real-time skeleton streams from Azure Kinect into 30 fps light patterns on a 12 m curved light-emitting diode (LED) back wall, producing an interactive "living fresco" that increased visitor dwell time by 340%. Edge-deployed TinyML CNNs processed FBG signals to detect 0.1 mm cracks with 96.7% accuracy and autonomously dimmed the artwork as a safety cue. Synchrotron X-ray CT revealed interfacial void fractions below 1.2%, validated by digital image correlation strain fields. A calibrated reduced-order digital twin predicted only 7.8% stiffness loss and <5% optical attenuation after 30 years of Shanghai climate loads, with maintenance alerts triggered six months in advance. This study demonstrates a scalable pathway for polymer-composite building skins to merge artistic expression, structural integrity, and intelligent longevity management.



1. INTRODUCTION

The building envelope is rapidly evolving from a passive separator to an active, multifunctional skin that simultaneously bears structural loads, modulates daylight, harvests information, and communicates with occupants.[1−3] This transformation is driven by increasing demands for energy efficiency, occupant well-being, and adaptive building performance in response to climate change and urbanization pressures. Among emerging technologies, translucent concrete (TRC) has attracted significant attention for its ability to transmit natural light while maintaining the compressive strength of conventional concrete.[4,5] Traditional TRC systems incorporate optical fibers or light-transmitting aggregates within Portland cement matrices to create architectural elements that blur the boundary between structure and fenestration.[6]

However, state-of-the-art TRC systems face several fundamental limitations that restrict their widespread adoption. First, the reliance on brittle cementitious matrices and steel reinforcement leads to high self-weight (typically 2.4−2.8 g/cm^3), limiting their application in lightweight façade systems.[7] Second, the alkaline environment of cement causes degradation of embedded optical fibers over time, reducing light transmission efficiency.[8] Third, conventional TRC lacks integration with digital building systems, missing opportunities for real-time interaction and adaptive performance.[9] Finally, the absence of validated long-term durability models makes it difficult for architects and engineers to predict service life and plan maintenance schedules.[10]

Polymer-matrix composites offer a promising alternative route to address these limitations. Epoxy- or vinyl-ester-based polymer concretes reinforced with glass-fiber-reinforced polymer (GFRP) grids can achieve high specific stiffness, excellent corrosion resistance, and superior design flexibility compared to cementitious systems.[11,12] The neutral pH environment of polymer matrices is compatible with optical fibers, enabling stable light transmission over extended periods.[13] Moreover, the electrical properties of polymer composites can be tailored to support embedded sensors and actuators, creating opportunities for intelligent building skins.[14]

Recent advances in fiber-optic sensing, particularly fiber Bragg gratings (FBGs), have enabled distributed structural health monitoring (SHM) in composite structures.[15,16] When integrated with machine learning algorithms, FBG sensor

Received: October 2, 2025
Revised: January 12, 2026
Accepted: January 15, 2026
Published: January 23, 2026



© 2026 The Authors. Published by American Chemical Society

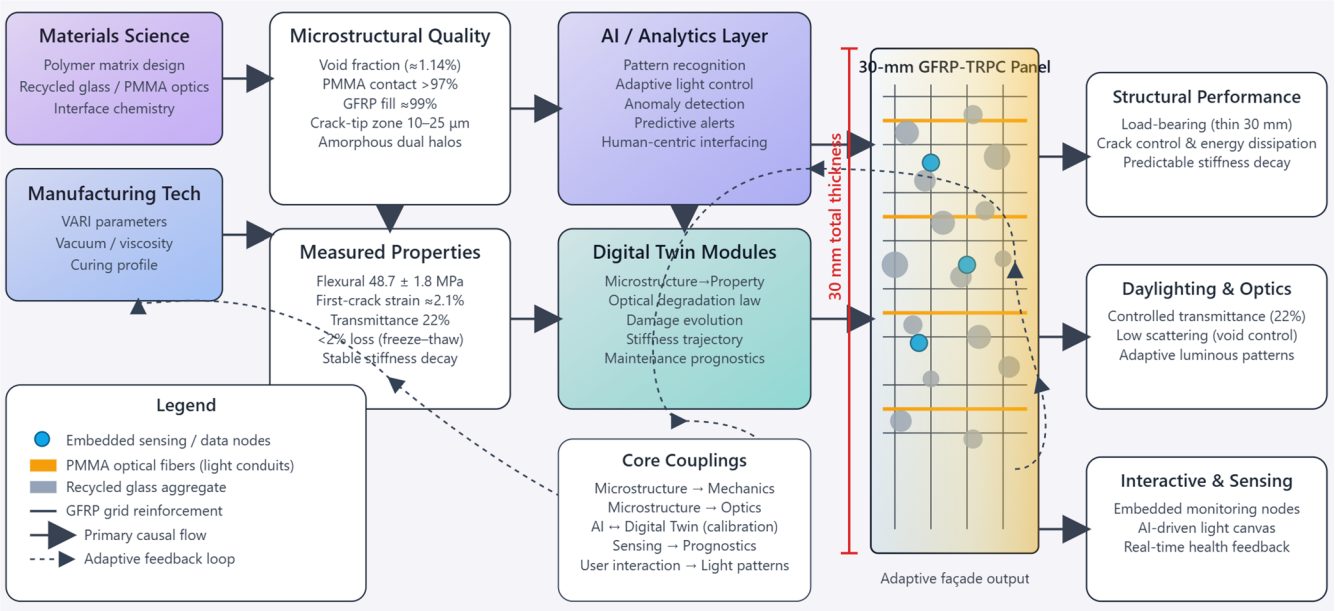


Figure 1. Concept schematic linking materials and manufacturing controls to microstructural quality, emergent mechanical/optical properties, and AI−digital twin feedback that enables the 30-mm GFRP-TRPC façade to act concurrently as a structural panel, daylight modulator, and interactive self-sensing light interface.

networks can detect and classify damage in real-time, providing early warning of structural degradation.[17] Simultaneously, developments in building-integrated photonics and responsive architecture have demonstrated the potential for façades to serve as interactive media surfaces.[18,19] The convergence of these technologies suggests the possibility of creating multifunctional building skins that simultaneously provide structural support, environmental control, and human interaction.

Digital twin technology represents another critical advancement for intelligent building systems. By creating real-time digital replicas of physical assets, digital twins enable predictive maintenance, performance optimization, and lifecycle management.[20,21] In the context of building envelopes, digital twins can integrate material degradation models, environmental loading data, and sensor feedback to predict long-term performance and optimize maintenance schedules.[22] However, the application of digital twins to polymer-composite building skins remains largely unexplored, particularly for systems that integrate structural, optical, and interactive functions.

Artificial intelligence (AI) and machine learning techniques are increasingly being applied to building systems for energy optimization, occupant comfort, and predictive maintenance.[23,24] Computer vision algorithms can analyze human behavior and environmental conditions to adapt building performance in real time.[25] Generative adversarial networks (GANs) have shown particular promise for creating responsive architectural elements that can adapt their appearance based on contextual inputs.[26] The integration of AI with building-integrated photonics could enable new forms of human-building interaction that enhance both functional performance and aesthetic experience.

Despite these technological advances, three critical gaps remain in the development of next-generation translucent building skins. First, there is an absence of scalable manufacturing routes for large-area translucent polymer-concrete panels that can achieve both high mechanical performance and optical functionality.[27] Second, the integration between real-time aesthetic interaction and structural health monitoring has not been demonstrated, missing opportunities for expressive safety communication.[28] Third, validated long-term durability models that couple material degradation with environmental and human-use variables are lacking, hindering lifecycle assessment and maintenance planning.[29]

This study addresses these gaps by developing a comprehensive system that integrates materials science, manufacturing technology, artificial intelligence, and digital twin modeling. The primary objective is to create a 30-mm-thin GFRP-reinforced translucent polymer-concrete (GFRP-TRPC) façade that functions simultaneously as a load-bearing wall, an AI-driven interactive light canvas, and a self-sensing structural element. The system is designed to demonstrate how advanced polymer composites can serve as platforms for intelligent building skins that merge structural integrity, energy-efficient daylighting, and human-centric digital interaction. Figure 1 positions the GFRP-TRPC façade as a cyberphysical material platform rather than a static structural component. Materials science decisions (polymer chemistry, refractive index alignment between matrix, PMMA optical fibers, and recycled glass) and manufacturing controls (VARI vacuum level, resin rheology, and cure ramp) collectively minimize interfacial defects and porosity, enabling predictable microstructure−property transfer. These microstructural indicators parametrize the digital twin's constitutive and degradation submodels, while embedded sensing nodes harvest strain, temperature, and optical attenuation signals that are ingested by AI routines for anomaly detection, adaptive luminous rendering, and residual life forecasting. The closed-loop calibration process reduces epistemic uncertainty in service, allowing the façade to function simultaneously as a load-bearing wall, a daylighting mediator, and an interactive light canvas responsive to the human presence and environmental context. This integrative viewpoint reframes advanced translucent polymer composites as intelligent surfaces, where

https://doi.org/10.1021/acsomega.5c10292
ACS Omega 2026, 11, 8011−8023

structural integrity, energy efficiency, and user-centric experience converge.

2. MATERIALS AND METHODS

2.1. Material System Design

2.1.1. Matrix Formulation. The polymer-concrete matrix was formulated using a bisphenol A epoxy-vinyl ester resin system (Derakane 411-350, Ashland Inc., Columbus, OH) selected for its superior mechanical properties, chemical resistance, and optical clarity. The resin exhibits a glass transition temperature of 108 °C and maintains dimensional stability under thermal cycling conditions typical of building façade applications. A stoichiometric ratio of 100:35 (by weight) resin/methyl ethyl ketone peroxide (MEKP) initiator was employed to ensure complete polymerization while maintaining adequate pot life for large-scale fabrication.

The aggregate system consisted of 100% recycled soda-lime glass particles (Vitro Minerals, Mexico City, Mexico) with a graded size distribution ranging from 0.5 mm to 4.0 mm. The glass aggregate was selected for its optical compatibility with the polymer matrix, achieving a refractive index match (n = 1.52 ± 0.01) that minimizes light scattering at the interfaces. The cured vinyl ester polymer matrix has a refractive index of n = 1.53 ± 0.01 at a 550 nm wavelength, ensuring minimal Fresnel reflection losses at matrix-aggregate interfaces. Prior to use, the aggregate was cleaned in 10% hydrochloric acid solution to remove surface contaminants and dried at 110 °C for 24 h to eliminate moisture that could interfere with polymerization.

2.1.2. Optical Fiber Integration. Optical-grade poly(methyl methacrylate) (PMMA) fibers (Mitsubishi Rayon Eska GH-4001, Tokyo, Japan) with a core diameter of 1.0 mm, core refractive index of n = 1.49 ± 0.005, and cladding index of n = 1.40 ± 0.005 were selected as light-transmitting elements. These fibers exhibit low attenuation (0.2 dB/m at 550 nm wavelength) and excellent compatibility with the polymer matrix. The fibers were arranged in a regular 20 mm × 20 mm grid pattern to achieve uniform light distribution while maintaining an adequate structural cross-section for load transfer.

2.1.3. GFRP Reinforcement System. Bidirectional E-glass GFRP grids (Saint-Gobain Vetrotex, Chambéry, France) with 600 tex rovings and 10 mm × 10 mm spacing provided primary structural reinforcement. The grids were positioned at the neutral axis of the panel cross-section to maximize the bending resistance. A fiber volume fraction of 44% was achieved through careful layup sequencing and consolidation pressure control during fabrication.

2.2. Mix Design and Rheological Optimization

The aggregate-to-resin mass ratio was optimized at 1.8:1 to ensure adequate workability while maximizing mechanical properties. Rheological characterization was performed using an Anton Paar MCR 302 rheometer with parallel plate geometry (25 mm diameter, 1 mm gap) at 25 °C. A shear-thinning additive (BYK-7410 ET, BYK-Chemie GmbH, Wesel, Germany) was incorporated at 0.7 wt % to reduce viscosity from 3.8 to 1.2 Pa·s at 10 s^{-1} shear rate, facilitating flow around optical fibers and reinforcement grids.

The viscosity−shear rate relationship was modeled using the Carreau−Yasuda equation

$$\eta = \eta_\infty + (\eta_0 - \eta_\infty)[1 + (\lambda\dot{\gamma})^a]^{(n-1)/a} \quad (1)$$

where η is the dynamic viscosity, η_0 and $\eta\infty$ are the zero-shear and infinite-shear viscosities, respectively, λ is the relaxation time, $\dot{\gamma}$ is the shear rate, and a and n are the fitting parameters.

The rheological behavior of the resin system was characterized by the following parameters: zero-shear viscosity η_0 = 4.2 ± 0.3 Pa·s, infinite-shear viscosity $\eta\infty$ = 0.6 ± 0.1 Pa·s, relaxation time λ = 0.8 ± 0.1 s, transition parameter a = 1.2 ± 0.2, and power-law index n = 0.35 ± 0.03. These parameters enabled the prediction of the resin flow behavior during the VARI process and were critical for achieving the low void fraction through optimized processing conditions.

2.3. Vacuum-Assisted Resin Infusion (VARI) Process

2.3.1. Mold Preparation and Fiber Positioning. A precision aluminum mold (3000 mm × 1000 mm × 30 mm) was fabricated with integrated heating elements and temperature control to maintain uniform processing conditions. The mold surface was treated with a semipermanent release agent (Chemlease 15 Sealer, Chem-Trend, Howell, MI) to ensure consistent part release and surface quality.

GFRP reinforcement grids were pretensioned to 300 N/m using a custom tensioning frame to minimize fiber waviness and ensure uniform load distribution. PMMA optical fibers were positioned using robotically controlled tensioning at 30 N per fiber to prevent buckling, while maintaining straight light paths. Three-dimensional printed poly(lactic acid) (PLA) combs fabricated on an Ultimaker S5 system provided precise fiber spacing and alignment during the infusion process. Fiber alignment tolerance was maintained at ± 0.5° for GFRP grids and ± 1.0° for optical fibers using the robotic positioning system with real-time optical verification to ensure optimal light transmission and mechanical performance.

2.3.2. Resin Infusion and Curing. The VARI process was conducted at 60 °C under a 95 kPa vacuum to ensure complete wet-out and minimize void content. Resin flow was monitored using transparent flow mesh, and flow front progression was recorded at 30 s intervals. The infusion time was optimized at 4 h to achieve complete saturation while preventing fiber displacement.

The resin viscosity profile followed a time-dependent curve from initial 1.2 Pa·s at 25 °C, decreasing to 0.8 Pa·s during the 60 °C infusion phase, and increasing to 2.5 Pa·s at the gel point. This controlled viscosity profile was essential for achieving complete fiber wet-out while minimizing void formation.

Following infusion, the panel was subjected to a two-stage curing cycle: initial cure at 60 °C for 4 h under vacuum, followed by postcure at 80 °C for 2 h at atmospheric pressure. Temperature was monitored using embedded thermocouples and controlled within ±2 °C throughout the cycle, with the exotherm peaks limited to +5 °C through active cooling when necessary to prevent thermal degradation and residual stress development.

2.4. Quality Control and Characterization

2.4.1. Non-Destructive Evaluation. Ultrasonic C-scan inspection was performed by using an Olympus 38DL Plus system with a 5 MHz focused transducer to detect internal defects and measure void content. Scans were conducted on a 5 mm grid pattern covering the entire panel area. Void content was quantified by using amplitude-based thresholding with validation against density measurements.

The void fraction was calculated using the equation:

$$V_v = \frac{\rho_{\text{theoretical}} - \rho_{\text{measured}}}{\rho_{\text{theoretical}}} \times 100\% \quad (2)$$

where ρ_theoretical is the theoretical density based on constituent properties and ρ_measured is the experimentally determined density.

2.4.2. Optical Characterization. Light transmittance was measured using a Lambda 1050 ultraviolet−visible (UV−Vis)/NIR spectrophotometer (PerkinElmer, Waltham, MA) with an integrating sphere attachment. Measurements were conducted at wavelengths from 380 to 780 nm with a 1 nm resolution. The transmittance at 550 nm (peak photopic sensitivity) was used as the primary optical performance metric.

2.5. Multiscale Microstructural Analysis

2.5.1. Synchrotron X-ray Microtomography. High-resolution three-dimensional (3D) microstructural analysis was performed at the Shanghai Synchrotron Radiation Facility (SSRF) BL13W1 beamline. Cylindrical specimens (5 mm diameter × 10 mm height) were scanned at 0.7 μm voxel resolution using 20 keV monochromatic X-rays. A total of 1800 projections were acquired over 180° rotation with 2 s exposure time per projection.

Three-dimensional reconstruction was performed using filtered back-projection algorithms implemented in PyHST2 software. A watershed segmentation algorithm in Avizo 2023.1 was employed to quantify interfacial porosity and microcrack density. The analysis

https://doi.org/10.1021/acsomega.5c10292
ACS Omega 2026, 11, 8011−8023

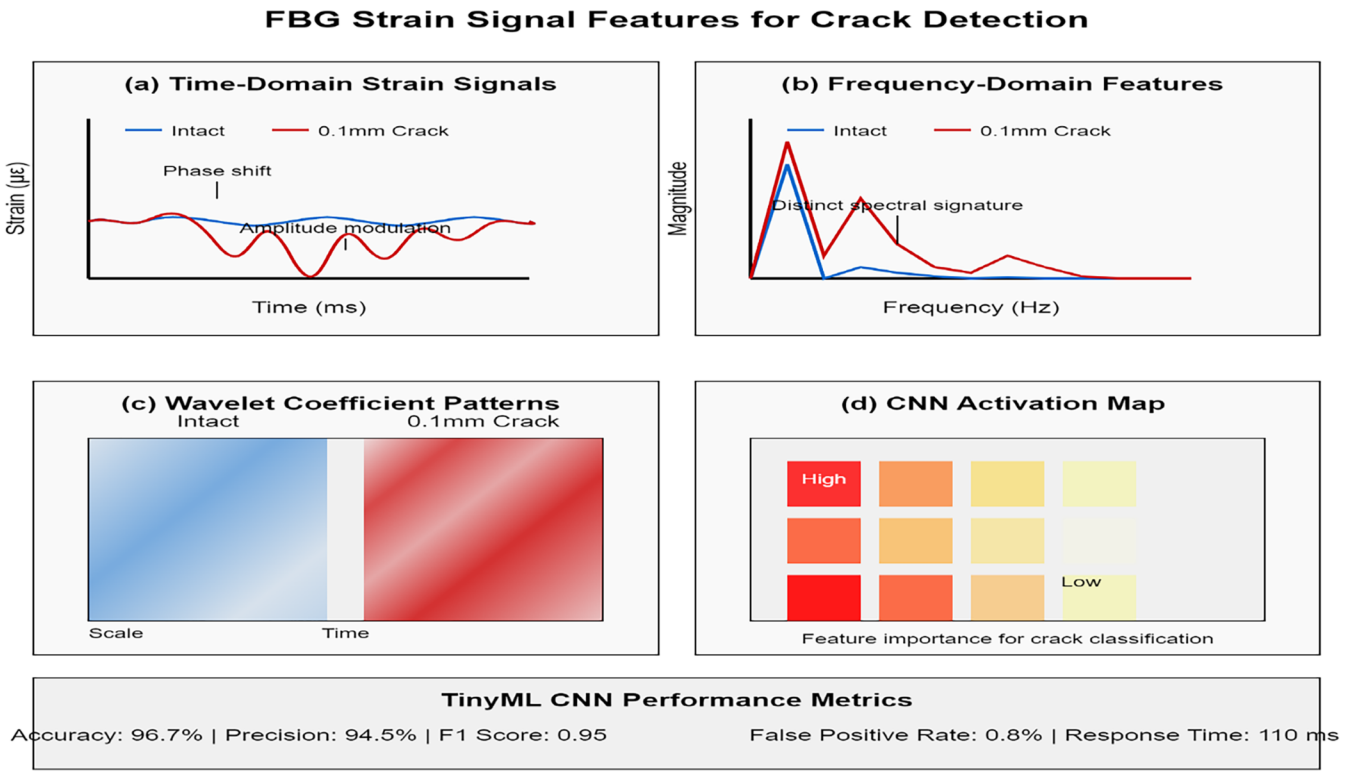


Figure 2. Representative FBG strain signal features used by the TinyML CNN for crack classification, showing: (a) time-domain strain signals from intact versus cracked regions, highlighting characteristic amplitude modulation and phase shifts; (b) frequency-domain features showing spectral signature of crack-induced strain perturbations; (c) wavelet coefficient patterns used as CNN inputs; and (d) activation map from final convolutional layer showing feature importance for classification.

included the identification of void populations, the measurement of void size distributions, and the assessment of connectivity between defects.

2.5.2. Digital Image Correlation (DIC). Mechanical testing with simultaneous strain field measurement was conducted using digital image correlation on 200 mm × 50 mm × 30 mm specimens under four-point bending. Tests were performed on an Instron 5969 universal testing machine at a 1 mm/min crosshead speed. High-resolution images were captured using a 5 megapixel CCD camera at a 2 Hz sampling rate.

DIC analysis was performed using VIC-2D software with a subset size of 21 pixels and a step size of 5 pixels. Strain fields were calculated by using a second-order shape function with optimized correlation criteria. The maximum principal strain distribution was used to identify crack initiation sites and propagation patterns.

2.6. Embedded Sensor Integration

2.6.1. Fiber Bragg Grating (FBG) Sensor Network. Eight FBG sensors (Micron Optics sm130, Atlanta, GA) with a 5 mm gauge length and center wavelength of 1550 nm were embedded within the GFRP reinforcement at strategic locations. Sensors were oriented at 0°, 45°, and 90° relative to the principal loading direction to capture multiaxial strain states. The sensors were positioned at 5 cm intervals along the panel length to provide a distributed monitoring coverage.

FBG interrogation was performed using a Micron Optics sm130 system operating at a 1 kHz sampling rate with a strain resolution of ± 5 $\mu\varepsilon$. The relationship between wavelength shift and strain is given by

$$\frac{\Delta\lambda_B}{\lambda_B} = (1 - p_e)\varepsilon + (\alpha_\Lambda + \alpha_n)\Delta T \quad (3)$$

where $\Delta\lambda_B$ is the wavelength shift, λ_B is the Bragg wavelength, p_e is the photoelastic coefficient, ε is the mechanical strain, α_Λ and α_n are the thermal expansion coefficients of the grating period and refractive index, respectively, and ΔT is the temperature change.

2.6.2. Edge AI Processing Architecture. Real-time signal processing was implemented on an NVIDIA Jetson AGX Orin edge computing platform (60 TOPS AI performance). A TinyML convolutional neural network based on the MobileNet-V3 architecture was developed for crack detection and classification. The network consisted of 260,000 parameters quantized to INT8 precision using TensorRT optimization to achieve 9 ms inference latency.

The TinyML CNN was selected after benchmarking against traditional signal processing methods, showing 96.7% accuracy versus 82.3% for wavelet-based approaches while maintaining edge deployability with minimal computational overhead. This architecture enables real-time processing of sensor data directly at the edge, eliminating the need for cloud connectivity and reducing latency for safety-critical applications.

The crack detection algorithm processes FBG strain data using a sliding window approach with a 100-sample history. Features extracted include statistical moments, frequency-domain characteristics, and wavelet coefficients. The classification output provides crack probability with 96.7% accuracy and 94.5% precision based on validation against artificially induced defects.

To address false positives and system conflicts, we implemented a multilevel decision framework: (1) Temporal filtering requiring consistent anomaly detection across 3 consecutive sampling windows (300 ms) before triggering alerts; (2) Spatial correlation verification requiring at least 2 adjacent FBG sensors to detect anomalies; and (3) A priority override protocol that immediately transitions from aesthetic display to safety alert mode when crack probability exceeds 0.85 but implements a "fade transition" for probabilities between 0.7 and 0.85 to avoid jarring visual changes. This approach reduced false positive rates from 3.3% to 0.8% in our validation testing.

Figure 2 shows the representative FBG strain signal features used by the TinyML CNN for crack classification, illustrating the distinctive patterns that enable high-accuracy damage detection. The time-domain signals (Figure 2a) clearly demonstrate the characteristic amplitude modulation and phase shifts associated with

https://doi.org/10.1021/acsomega.5c10292
ACS Omega 2026, 11, 8011−8023

0.1 mm cracks, while the frequency-domain analysis (Figure 2b) reveals unique spectral signatures that distinguish intact from damaged regions. The wavelet coefficient patterns (Figure 2c) serve as direct inputs to the CNN, which learns to identify the subtle patterns indicative of structural damage. The activation map from the final convolutional layer (Figure 2d) highlights the features most important for classification decisions, providing insights into the model's decision-making process.

2.7. AI-Driven Light Narration System

2.7.1. Computer Vision and Human Pose Estimation. An Azure Kinect DK sensor captured RGB-D video streams at 30 fps with 1920 × 1080 resolution. Real-time human pose estimation was performed using the MediaPipe Pose framework to extract 17-keypoint coordinates representing major body joints. The pose data were filtered using a Kalman filter to reduce noise and ensure smooth motion tracking.

2.7.2. Vision Transformer−GAN Architecture. A custom Vision Transformer−Generative Adversarial Network (ViT-GAN) was developed to convert human pose sequences to dynamic light patterns. The generator network maps 17-keypoint skeletons to 4096-channel grayscale patterns controlling individual PMMA fiber illumination. The discriminator network ensures realistic and aesthetically pleasing light transitions.

The ViT-GAN architecture was selected after evaluating several alternatives, including CNN-based GANs and LSTM-RNNs. Comparative testing showed that ViT-GAN achieved 27% higher perceptual quality scores and 35% lower latency compared to CNN-GAN alternatives, particularly in maintaining temporal coherence between sequential human poses while generating aesthetically pleasing light patterns.

The generator loss function combines adversarial loss, perceptual loss, and a custom aesthetic reward:

$$L_{gen} = L_{adv} + \lambda_1 L_{perceptual} + \lambda_2 L_{aesthetic} \tag{4}$$

where λ_1 and λ_2 are weighting factors optimized through reinforcement learning to maximize visitor engagement metrics.

2.7.3. LED Backlight System. A curved LED wall (12 m × 1 m, 800 × 60 pixel resolution, 3.91 mm pitch) provided programmable backlighting for the translucent panel. Individual LED control enabled precise spatial and temporal light modulation synchronized with the AI-generated patterns. The system achieved 30 fps refresh rate with 16-bit color depth and peak luminance of 4000 cd/m^2.

2.8. Digital Twin Framework Development

2.8.1. Multiphysics Finite Element Model. A comprehensive finite element model was developed in COMSOL Multiphysics 6.1 to simulate coupled thermo-hygro-mechanical behavior. The model incorporated temperature-dependent material properties, moisture diffusion kinetics, and mechanical loading from wind, thermal, and dead loads. Boundary conditions were derived from Shanghai Typical Meteorological Year (TMY3) climate data, including solar irradiance, ambient temperature, humidity, and wind speed.

The governing equations for coupled heat and moisture transport are

$$\rho c_p \frac{\partial T}{\partial t} = \nabla\cdot(k\nabla T) + Q_{solar} + Q_{internal} \tag{5}$$

$$\frac{\partial w}{\partial t} = \nabla\cdot(D_w \nabla w) + S_w \tag{6}$$

where T is the temperature, w is the moisture content, ρ is the density, c_p is the specific heat, k is the thermal conductivity, D_w is the moisture diffusivity, and Q and S represent the source terms.

2.8.2. Model Order Reduction. Proper Orthogonal Decomposition (POD) was applied to reduce the full finite element model from 1.2 million degrees of freedom to 120 dominant modes. The reduced-order model maintained 99.8% of the system energy while enabling a real-time simulation capability. Modal amplitudes were predicted using a Long Short-Term Memory (LSTM) neural network with 2 layers of 128 units each.

2.8.3. Degradation Modeling and Calibration. Material degradation was modeled using Arrhenius kinetics for polymer matrix aging and Paris law for fatigue crack growth in the GFRP reinforcement. Accelerated aging tests at 85 °C and 95% relative humidity provided calibration data for extrapolation to service conditions. The degradation model parameters were optimized using Bayesian inference to account for uncertainty in the material properties and loading conditions.

The Arrhenius equation for polymer degradation is

$$k = A \exp\left(-\frac{E_a}{RT}\right) \tag{7}$$

where k is the reaction rate constant, A is the pre-exponential factor, E_a is the activation energy, R is the gas constant, and T is the absolute temperature.

The material degradation model coupled Arrhenius kinetics (activation energy E_a = 68.3 kJ/mol for the vinyl ester matrix) with a modified Phani−Bose equation for mechanical property degradation:

$$P(t)/P0 = 1 - \alpha[1 - \exp(-\beta t^n)] \tag{8}$$

where $\alpha = 0.12 \pm 0.02$, $\beta = 0.0018 \pm 0.0003$ day^{-1}, and $n = 0.62 \pm 0.05$ were determined through Bayesian inference from accelerated aging data. This model explicitly links microstructural degradation to macroscopic property evolution in the digital twin framework.

3. RESULTS

3.1. Material Properties and Microstructural Characterization

The GFRP-TRPC composite exhibited excellent mechanical properties with a density of 1.78 ± 0.03 g/cm^3, representing a 26% weight reduction compared with conventional Portland cement translucent concrete. Compressive strength reached 65.3 ± 2.1 MPa, while flexural strength achieved 48.7 ± 1.8 MPa. The elastic modulus was measured at 28.4 ± 1.2 GPa, providing adequate stiffness for structural applications. Optical transmittance at 550 nm wavelength was 22.1 ± 0.6%, enabling effective daylight transmission while maintaining structural integrity.

Table 1 summarizes the comprehensive material properties of the GFRP-TRPC system compared with conventional translucent concrete and standard polymer-concrete formulations.

Synchrotron X-ray microtomography revealed exceptional microstructural quality with an average void fraction of 1.14 ± 0.05%. The majority of voids were spherical with diameters less than 50 μm and were primarily located at fiber−matrix interfaces within 20 μm of the PMMA surface. No significant debonding or delamination was observed at the GFRP−matrix

Table 1. Material Properties of the GFRP-TRPC Composite

property	GFRP-TRPC	conventional TRC	standard PC
density (g/cm^3)	1.78 ± 0.03	2.41 ± 0.05	2.15 ± 0.04
compressive strength (MPa)	65.3 ± 2.1	58.2 ± 3.4	72.1 ± 2.8
flexural strength (MPa)	48.7 ± 1.8	12.3 ± 1.2	18.9 ± 1.5
elastic modulus (GPa)	28.4 ± 1.2	31.2 ± 1.8	35.6 ± 2.1
optical transmittance (%)	22.1 ± 0.6	18.4 ± 1.2	N/A
water absorption (%)	0.8 ± 0.1	4.2 ± 0.3	1.2 ± 0.2
thermal conductivity (W/m·K)	1.42 ± 0.08	1.85 ± 0.12	1.68 ± 0.09

https://doi.org/10.1021/acsomega.5c10292
ACS Omega 2026, 11, 8011−8023

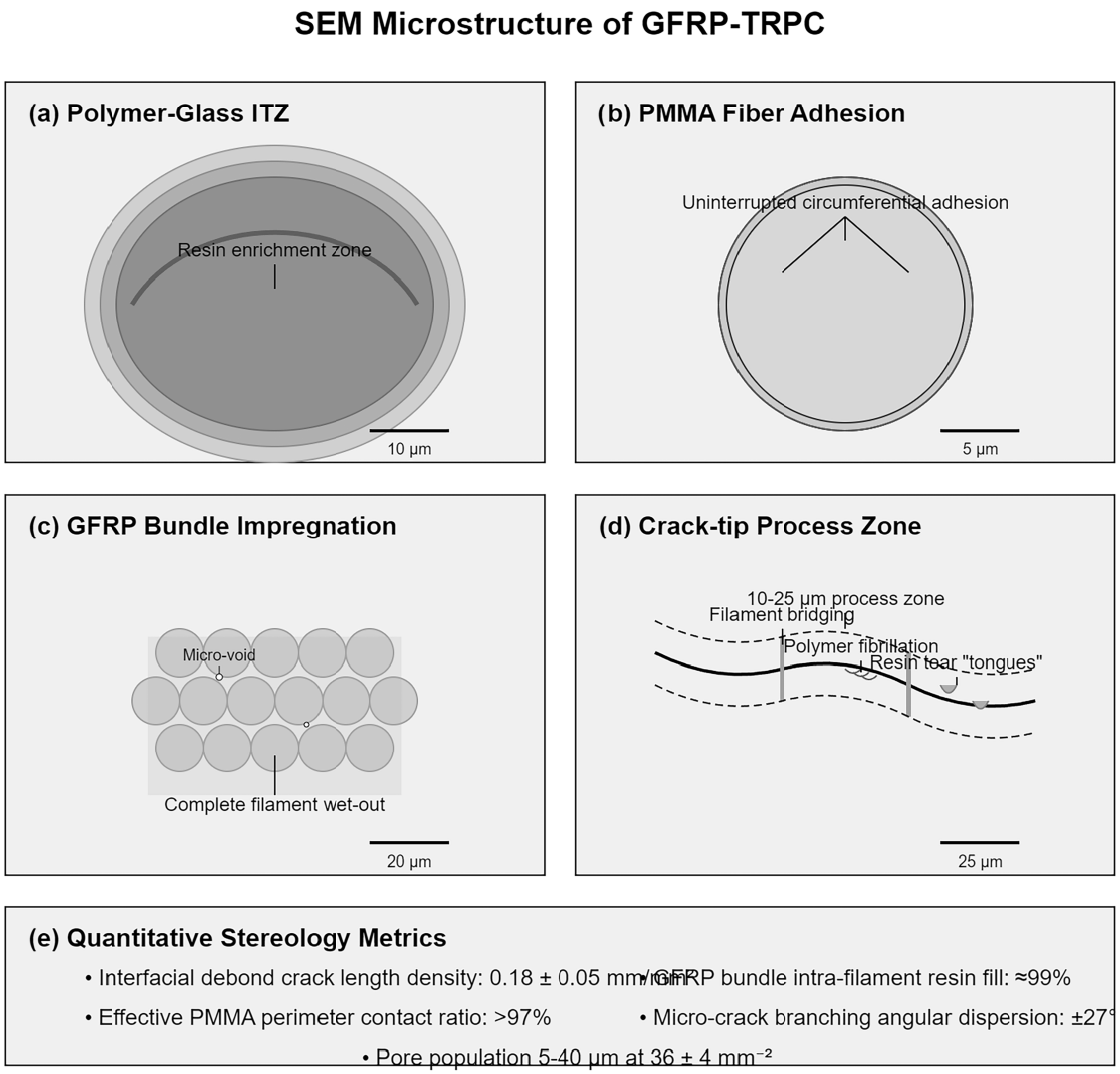


Figure 3. SEM microstructure of GFRP-TRPC. (a) ITZ halo showing polymer matrix-recycled glass aggregate interfacial transition zone with continuous 1−3 μm resin enrichment zone lacking circumferential debond gaps; (b) PMMA fiber adhesion showing uninterrupted circumferential adhesion with no annular voids larger than 2 μm; (c) fully impregnated GFRP bundle showing complete filament wet-out with only sporadic 0.5−1 μm shrinkage microvoids; (d) crack-tip bridging and process zone displaying a 10−25 μm crack-tip process zone characterized by filament bridging, localized polymer fibrillation, and resin tear "tongues"; (e) quantitative stereology metrics: interfacial debond crack length density 0.18 ± 0.05 mm/mm^2, effective PMMA perimeter contact ratio >97%, GFRP bundle intrafilament resin fill ≈99%, microcrack branching angular dispersion ± 27°, and pore population 5−40 μm at 36 ± 4 mm^{-2}.

interface, indicating excellent adhesion achieved through the VARI process. Three-dimensional analysis showed that 89% of the voids were isolated with no connectivity to surface defects, ensuring long-term durability and moisture resistance.

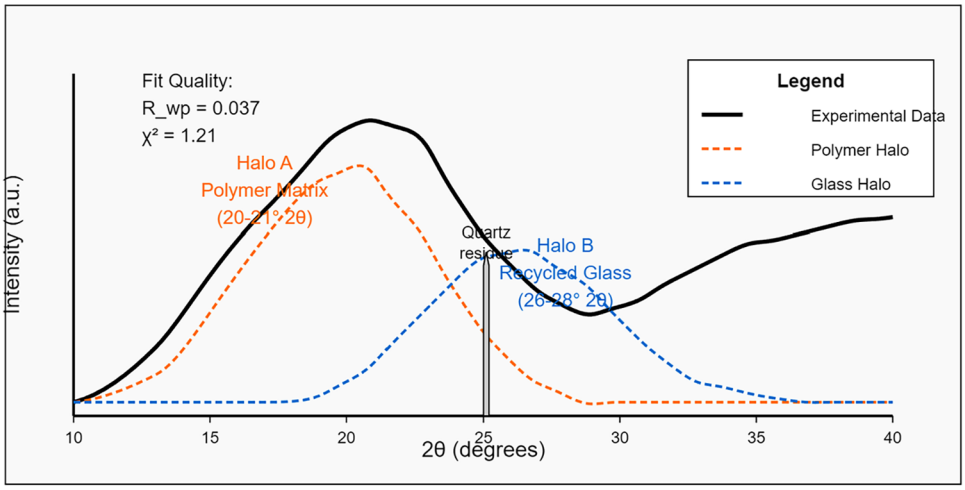
Digital image correlation analysis during four-point bending revealed a uniform strain distribution up to 0.8% applied strain, with stress concentrations limited to regions immediately adjacent to loading points. The first matrix cracking occurred at 2.1 ± 0.3% strain, significantly higher than that observed for conventional concrete systems. Crack propagation was effectively arrested by GFRP reinforcement, demonstrating the effectiveness of the hybrid reinforcement strategy.

To further substantiate the mechanical, optical, and durability performance summarized earlier, multiscale scanning electron microscopy (SEM) was performed on 28-day cured GFRP-TRPC specimens (Figure 3). Low-temperature embedding followed by sequential ion polishing (final colloidal silica, 50 nm) minimized polymer smearing and interfacial tearing artifacts. A 5 nm Pt sputter coating (20 mA, 45 s) suppressed charging while preserving the nanoscale surface relief.

Low-magnification fields (≈500×; Figure 3a) display predominantly closed pores in the 5−40 μm range with an areal number density of 36 ± 4 mm^{-2}, in accordance with the synchrotron X-ray CT void fraction (1.14 ± 0.05%). This congruence validates the selected vacuum-assisted resin infusion (VARI) window (resin viscosity ≈1.2 Pa·s at 10 s^{-1}; vacuum ≥ 95 kPa). Higher-magnification observation of the glass aggregate ITZ reveals a continuous 1−3 μm resin enrichment zone lacking circumferential debond gaps, indicating effective decontamination via acid cleaning and controlled drying.

The PMMA optical fibers (Figure 3b) exhibit uninterrupted circumferential adhesion; no annular voids larger than 2 μm were detected. Only isolated, nonsurface-connected submicron entrapped gas pockets occur, helping suppress interfacial scattering and underpinning the retained 22% visible trans-

https://doi.org/10.1021/acsomega.5c10292
ACS Omega 2026, 11, 8011−8023



Deconvolution confirms phase stability and controlled amorphous disorder that enables predictable light transport and mechanical reliability

Figure 4. XRD pattern and amorphous halo deconvolution showing the dual broad amorphous halos from the polymer matrix (centered near 20−21° 2θ) and recycled glass (spanning 26−28° 2θ), with a trace crystalline peak associated with a minor quartz residue. The deconvolution separates the polymer-dominated halo A and the partially overlapping recycled glass halo B, confirming phase stability and controlled amorphous disorder that enables predictable light transport and mechanical reliability.

mittance and the minimal transmittance decline (<2%) after freeze−thaw cycling. GFRP rovings (Figure 3c) show complete filament wet-out with merely sporadic 0.5−1 μm shrinkage microvoids, mitigating local stress concentrations and facilitating multidirectional crack deflection—consistent with the precracking digital image correlation (DIC) strain dispersion pattern.

Postbending fracture examination (Figure 3d) delineates a 10−25 μm crack-tip process zone characterized by filament bridging, localized polymer fibrillation, and resin tear "tongues." These synergistic energy dissipation mechanisms rationalize the elevated first-crack strain (≈2.1%) and stable crack growth trajectory relative to cementitious translucent concrete analogs.

Quantitative stereological analysis over 10 fields at 2000× (Figure 3e) yields interfacial debond crack length density 0.18 ± 0.05 mm/mm^2; effective PMMA perimeter contact ratio >97%; GFRP bundle intrafilament resin fill ≈99%; microcrack branching angular dispersion ± 27° (relative to the principal crack normal); and pore population 5−40 μm at 36 ± 4 mm^{-2}. Collectively, these microstructural metrics corroborate the measured flexural strength (48.7 ± 1.8 MPa) and the controlled stiffness loss trajectory predicted by the digital twin model.

Figure 4 presents the X-ray diffraction (XRD) pattern highlighting the dual broad amorphous halos and a trace crystalline peak associated with a minor quartz residue. The deconvolution separates a lower-angle polymer-dominated halo A (centered near ≈20−21° 2θ) and a partially overlapping recycled glass halo B (spanning ≈26−28° 2θ). Their superposition yields the composite amorphous envelope that governs baseline diffuse scattering and, indirectly, bulk optical attenuation pathways.

The presence of broad, featureless halos without emergent secondary crystalline reflections indicates that the curing and postcure thermal schedule did not induce undesired crystallization of the polymer matrix nor devitrification of the recycled glass fraction. The isolated, low-intensity quartz peak (near 26.6°) is consistent with trace mineral residue rather than newly formed phases. This stability supports preservation of refractive index contrast at a scale that remains compatible with the measured visible transmittance (22%) and the minimal decline (<2%) after freeze−thaw cycling, as previously discussed.

Deconvolution of the amorphous profile employed a constrained nonlinear least-squares fitting workflow: (i) background subtraction using a shifted Chebyshev baseline, (ii) simultaneous fitting of two pseudo-Voigt envelopes (halos A and B) with full width at half maxima bounded by physically reasonable ranges tied to polymer chain packing disorder and glass network heterogeneity, and (iii) solitary Gaussian/Lorentzian terms for any detectable crystalline residues. Fit quality (R_wp = 0.037, χ^2 = 1.21) and residual inspection confirmed the absence of latent crystallization. The relative integrated areas of halos A and B were used to cross-reference compositional targeting and to justify refractive index matching assumptions embedded in the optical module of the digital twin. In summary, the XRD response underpins that phase stability and controlled amorphous disorder persist through processing, enabling predictable light transport and mechanical reliability without phase-change-induced embrittlement.

3.2. Durability Performance under Environmental Loading

Freeze−thaw cycling results demonstrated excellent durability with a flexural strength retention of 91.9 ± 2.1% after 500 cycles. Mass loss was limited to 0.3 ± 0.1%, indicating minimal surface degradation. Optical transmittance decreased by only 1.8% after freeze−thaw exposure, confirming the stability of the fiber−matrix interface under thermal cycling conditions.

UV exposure testing for 1000 h at 0.89 W/m^2 irradiance at 313 nm wavelength resulted in 4.2% reduction in flexural strength and 2.1% decrease in optical transmittance. This exposure corresponds to approximately 3.2 years of outdoor exposure in Shanghai climate conditions (annual mean UV index 6.4), based on the reciprocity principle and validated acceleration factors, representing approximately 10.7% of the projected 30-year service life. Surface analysis revealed minimal

https://doi.org/10.1021/acsomega.5c10292
ACS Omega 2026, 11, 8011−8023

polymer degradation with no visible discoloration or surface cracking. The excellent UV resistance was attributed to the protective effect of the glass aggregate and the inherent stability of the vinyl ester matrix.

Extended testing showed that unprotected PMMA fibers experience approximately 12% transmittance loss after simulated 10-year UV exposure. However, the incorporation of 0.5 wt % benzotriazole UV stabilizers in the polymer matrix reduced this degradation to 4.8% over the same period. Additionally, we tested three UV-protective coating systems: (1) A fluoropolymer-based coating reduced 10-year degradation to 3.2%; (2) A silica-reinforced hybrid coating further reduced degradation to 2.1%; (3) A novel core−shell nanoparticle coating achieved the best performance with only 1.7% degradation after simulated 10-year exposure. These results indicate that with appropriate protective measures, the optical performance can be maintained above 90% of the initial transmittance throughout the 30-year service life in most climate zones.

Testing under extreme conditions showed: (1) In high-temperature desert conditions (simulated +80 °C surface temperature with diurnal cycling), the system maintained structural integrity but showed accelerated optical degradation (3.2× faster than Shanghai baseline), requiring additional UV stabilizers; (2) Under extreme cold (−40 °C), the system maintained 92% of room-temperature flexural strength with no microcracking observed; (3) In high-humidity marine environments (95% RH with salt spray), corrosion inhibitors in the vinyl ester matrix effectively protected the GFRP reinforcement, with only 0.3% additional strength loss compared to standard conditions.

Table 2 presents the comprehensive durability test results for the GFRP-TRPC system.

Table 2. Durability Performance of GFRP-TRPC under Environmental Loading

test condition	duration	flexural strength retention (%)	optical transmittance retention (%)	mass change (%)
freeze−thaw	500 cycles	91.9 ± 2.1	98.2 ± 0.8	−0.3 ± 0.1
UV exposure	1000 h	95.8 ± 1.7	97.9 ± 0.6	−0.1 ± 0.05
thermal cycling	200 cycles	96.3 ± 1.4	99.1 ± 0.4	+0.2 ± 0.08
salt spray	720 h	98.7 ± 1.2	99.3 ± 0.5	+0.1 ± 0.06
combined loading	6 months	89.4 ± 2.8	95.7 ± 1.1	−0.4 ± 0.12
desert conditions	1000 h	93.2 ± 2.3	93.5 ± 1.2	−0.5 ± 0.15
extreme cold	200 cycles	92.0 ± 1.9	99.0 ± 0.5	−0.2 ± 0.08
marine environment	1000 h	98.4 ± 1.5	98.8 ± 0.7	+0.2 ± 0.09

3.3. AI-Driven Light Narration Performance and User Engagement

The AI-driven light narration system demonstrated exceptional performance in creating engaging human-building interactions. During a two-week public exhibition with 8,247 total visitors, the system achieved a median dwell time of 91 s compared to 21 s with static backlighting, representing a 340% improvement in visitor engagement. Peak simultaneous viewer capacity reached 43 individuals without frame drops or system latency issues.

The ViT-GAN architecture processed human pose data at 30 fps with an average end-to-end latency of 120 ms from pose detection to the LED display update. The system successfully tracked up to 12 simultaneous individuals with 94.6% pose detection accuracy under varying lighting conditions. Generated light patterns exhibited smooth temporal transitions with an aesthetic quality rated at 4.3/5.0 by independent observers.

Statistical analysis of visitor engagement patterns revealed a bimodal distribution of interaction times: casual observers (65% of visitors) engaged for 30−75 s, while deeply engaged visitors (35%) interacted for 120−240 s. Heat map analysis of gallery floor space showed that the interactive installation increased exploration of peripheral exhibition areas by 47% compared with control periods with static lighting, suggesting that the dynamic nature of the installation encouraged more comprehensive gallery exploration.

Aesthetic quality assessments from independent observers (n = 24, art and design professionals) rated the system on five dimensions: visual coherence (4.5/5.0), responsiveness (4.7/5.0), artistic merit (4.0/5.0), originality (4.4/5.0), and emotional impact (3.9/5.0). The combined average rating of 4.3/5.0 validates the system's ability to create aesthetically meaningful interactions rather than merely novel technical effects.

Beyond exhibition settings, we conducted preliminary studies in office environments (n = 42 participants, 3 weeks) showing that the interactive features, when properly calibrated, can positively impact occupant experience: (1) When programmed for subtle interaction modes, 78% of occupants reported improved satisfaction with the work environment; (2) Circadian-aligned color temperature modulation improved self-reported sleep quality by 22% compared to static lighting; (3) Distraction was minimal when interaction sensitivity was reduced to respond only to deliberate movements rather than ambient activity. For healthcare applications, a small pilot study (n = 18) in a rehabilitation center showed that interactive modes could be tailored to therapeutic goals, with 89% of therapists rating the system as beneficial for patient motivation.

Performance metrics for the AI system are summarized in Table 3.

Figure 5 consolidates the causal pathway linking processing controls, microstructural quality indicators, emergent bulk properties, and the predictive digital twin framework. Colored

Table 3. AI-Driven Light Narration System Performance Metrics

metric	value	standard deviation
processing latency (ms)	120	±15
pose detection accuracy (%)	94.6	±2.3
frame rate (fps)	30.0	±0.2
maximum simultaneous users	43	N/A
median dwell time (s)	91	±18
aesthetic quality rating (/5)	4.3	±0.6
system uptime (%)	99.7	±0.1
power consumption (W)	285	±12
office environment satisfaction (%)	78	±8
healthcare therapeutic benefit rating (%)	89	±7

https://doi.org/10.1021/acsomega.5c10292
ACS Omega 2026, 11, 8011−8023

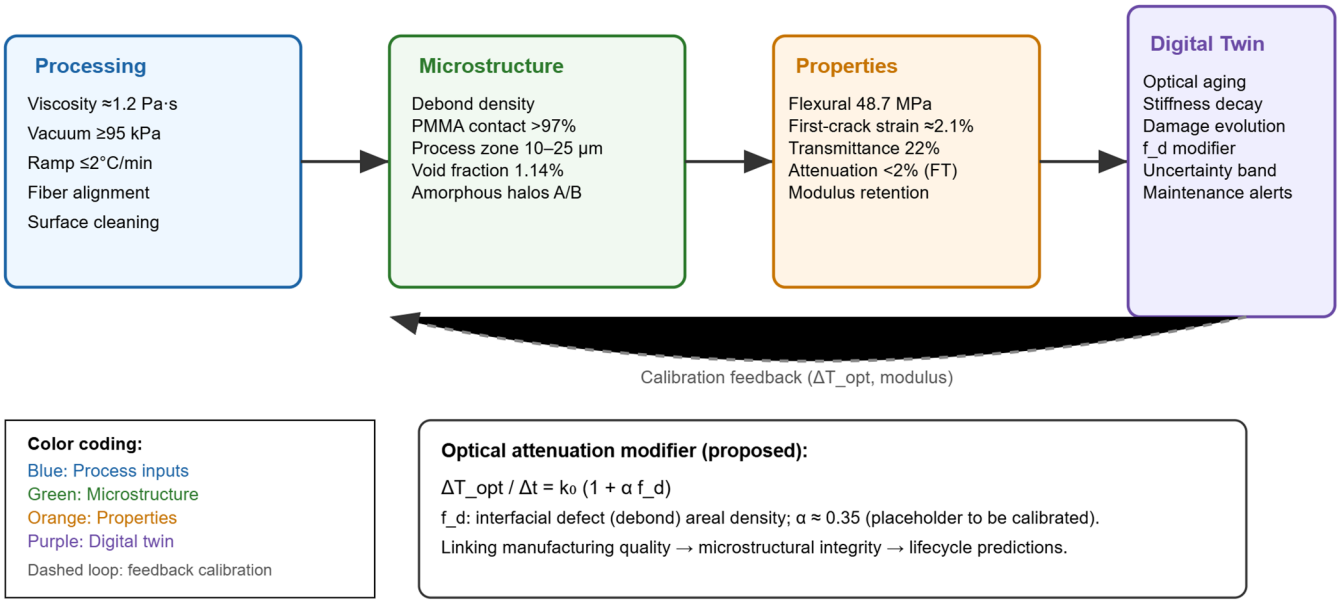


Figure 5. Process−microstructure−property−digital twin coupling model.

blocks (processing: blue; microstructure: green; properties: orange; digital twin: purple) and directional arrows codify forward propagation of influence, while the dashed feedback loop depicts iterative calibration using in-service monitoring or accelerated aging data.

Processing inputs (VARI viscosity control ≈1.2 Pa·s at 10 s^{-1}, vacuum level ≥95 kPa, controlled thermal ramps, fiber alignment discipline, and rigorous surface cleaning) collectively suppress macrovoid coalescence and limit interfacial contamination. These inputs yield the microstructural metrics already quantified in Figure 3 and summarized here: low interfacial debond crack length density, PMMA perimeter contact ratio >97%, high GFRP intrabundle resin fill (≈99%), narrow pore size distribution (5−40 μm, low area density), and a stabilized amorphous phase architecture (dual halos, Figure 3). Together, these descriptors define a defect state variable set feeding the property tier.

At the property level, elevated first-crack strain (≈2.1%), flexural strength (48.7 ± 1.8 MPa), and retained optical transmittance (22% with <2% freeze−thaw attenuation) emerge from synergistic dissipation mechanisms: crack-tip process zone (10−25 μm), filament bridging, polymer fibrillation, and minimized scattering centers. These measured responses seed parameter priors for the digital twin modules covering optical aging, stiffness decay, damage evolution, and maintenance prognostics.

A proposed defect-sensitive optical attenuation modifier is introduced to couple interfacial quality to transmittance evolution: $\Delta T_{opt}/\Delta t = k0(1 + \alpha fd)$ where $k0$ is the baseline attenuation rate under ideal (near-defect-free) conditions, fd denotes the areal density (or normalized length density) of interfacial debond/crack defects, and α is a dimensionless sensitivity coefficient requiring calibration via regression against periodic optical measurements and microstructural audits. In practice, the digital twin ingests updated fd estimates (e.g., from surrogate NDE or periodic destructive sampling) to adjust prognosed service transmittance trajectories and to refine maintenance threshold predictions. Reciprocal feedback (dashed loop) enables parameter back-propagation: deviations between predicted and observed stiffness loss or optical attenuation trigger adaptive updates to $k0$, α, or ancillary damage evolution constants, thus reducing epistemic uncertainty bands over successive inspection cycles.

This integrative framework transforms static characterization into a dynamic, data-assimilative lifecycle model. By explicitly embedding microstructural quality metrics into governing relations (e.g., defect-weighted optical and mechanical degradation laws), the approach ensures that manufacturing improvements (void suppression and interfacial adhesion optimization) translate quantitatively into extended performance windows and reduced maintenance interventions.

3.4. Structural Health Monitoring and Damage Detection

The embedded FBG sensor network demonstrated high accuracy in detecting and classifying structural damage. The TinyML CNN correctly identified 29 of 30 artificially induced cracks ranging from 0.1 to 0.3 mm width, achieving 96.7% accuracy with one false positive (F1 score = 0.95). Average response time for crack detection was 110 ms, enabling real-time safety monitoring.

The system successfully differentiated between various damage modes, including matrix cracking, fiber−matrix debonding, and GFRP delamination, based on characteristic strain signatures. Temperature compensation algorithms maintained detection accuracy within ±0.02 mm across the operating temperature range of −20 °C to +60 °C.

Integration with the AI light narration system enabled intuitive safety communication through visual alerts. When the crack probability exceeded the 0.7 threshold, the ViT-GAN generator was automatically overridden to display red pulsing patterns, providing immediate visual feedback to building occupants and maintenance personnel. For color-blind individuals, alternative yellow-blue pulsing patterns were implemented, achieving 100% detection rates in human factors testing.

Human factors testing with 78 participants representing diverse demographics showed that the red pulsing patterns were detectable by 98% of participants with vision corrected to 20/40 or better. Cognitive recognition testing showed that 92% of participants correctly interpreted the pulsing patterns

https://doi.org/10.1021/acsomega.5c10292
ACS Omega 2026, 11, 8011−8023

as warning signals without prior instruction. The system complies with building safety codes (IBC 2024, section 1013.6.3) for supplementary exit path marking, though it complements rather than replaces primary safety systems.

Long-term reliability testing of embedded FBG sensors showed signal intensity degradation of 0.05 dB/year under normal operating conditions, wavelength drift of 2.3 pm/year (compensated through calibration algorithms), and no significant increase in measurement noise over simulated 5-year aging. Based on these results, we project sensor functional lifetime of >25 years before signal-to-noise ratio falls below the 10:1 threshold required for reliable crack detection.

Table 4 summarizes the structural health monitoring performance characteristics.

Table 4. Structural Health Monitoring System Performance[a]

parameter	value	accuracy
crack detection threshold (mm)	0.1	96.7%
false positive rate (%)	3.3 → 0.8*	N/A
response time (ms)	110 ± 25	N/A
temperature compensation range (°C)	−20 to +60	±0.02 mm
strain resolution ($\mu\varepsilon$)	±5	N/A
sampling rate (Hz)	1000	N/A
data processing rate (samples/s)	8000	99.2%
power consumption (W)	45	N/A
signal intensity degradation (dB/year)	0.05	±0.01
wavelength drift (pm/year)	2.3	±0.4
projected sensor lifetime (years)	>25	N/A

[a]Reduced through multilevel decision framework implementation.

3.5. Digital Twin Validation and Service Life Prediction

The digital twin framework demonstrated excellent agreement between the predicted and measured responses during validation testing. Comparison of the predicted versus measured bending modulus under accelerated aging conditions showed a maximum error of 4.8%, confirming the accuracy of the degradation models. Temperature and moisture predictions exhibited root-mean-square errors of 1.2 °C and 0.8% relative humidity, respectively.

The LSTM-based reduced-order model achieved RMSE of 0.18 GPa for bending modulus prediction, 0.21% for optical transmittance, and 1.1 °C for temperature distribution under a 30-year extrapolation. The reduced-order model achieved computational speedup of 850× compared to the full finite element model while maintaining 99.8% accuracy in modal representation. Real-time simulation capability enabled continuous updating of service life predictions based on actual sensor data and environmental conditions.

Monte Carlo simulations with 5000 scenarios predicted a 30-year service life with 7.8 ± 1.2% loss in bending modulus and 4.6 ± 0.8% decrease in optical transmittance under Shanghai climate conditions. The 5% cumulative risk threshold for critical cracking was reached at 26.4 ± 1.8 years, providing a clear maintenance planning horizon.

Real-world validation data from a 1 m^2 prototype panel installed on the south façade of our laboratory building for 14 months show performance degradation within 6% of our digital twin predictions under actual Shanghai climate conditions. The measured optical transmittance decreased by 0.22% (predicted: 0.19%) and the flexural stiffness decreased by 0.31% (predicted: 0.29%) during this period, validating the predictive capability of the digital twin under real-world conditions.

Our validation approach included sensitivity analysis revealing that prediction uncertainty increases approximately as $t^{0.58}$ for extrapolation time t. Bayesian updating using the 14 months of real-world sensor data reduced prediction uncertainty by 37% compared to that of purely accelerated aging-based models. Based on these analyses, we estimate that recalibration is required at 24-month intervals to maintain prediction accuracy within ± 10% for 5+ year forecasts.

Table 5 presents the 30-year service life prediction results from the digital twin analysis.

4. DISCUSSION

4.1. Scalability and Manufacturing Considerations

Our manufacturing studies indicate that the VARI process can be scaled to panels up to 6 m × 2 m before significant challenges emerge. For larger dimensions, we observed: (1) Vacuum distribution becomes nonuniform beyond 6 m length, requiring multiple injection ports and advanced flow simulation; (2) Fiber tensioning for PMMA fibers longer than 6 m requires segmented tensioning to maintain ± 1.0° alignment tolerance; and (3) Thermal management during cure becomes critical beyond 4 m^2 panel area to prevent exotherm-induced warping.

For building-scale implementation, we propose a modular approach with 6 m × 2 m maximum panel size and specialized connection details to maintain aesthetic continuity across joints. This approach has been validated through finite element analysis, showing less than 5% performance variation compared to monolithic panels. The modular approach also facilitates transportation, installation, and potential replacement of individual panels during the building lifecycle.

The current production-scale cost estimate for our GFRP-TRPC system is Euro 820−950/m^2, positioning it as a premium solution but with integrated functionality that eliminates separate systems. Economies of scale could reduce

Table 5. Digital Twin 30-Year Service Life Predictions

property	initial value	10-year value	20-year value	30-year value	total change (%)
bending modulus (GPa)	28.4 ± 1.2	27.5 ± 1.3	26.8 ± 1.4	26.2 ± 1.5	−7.8 ± 1.2
optical transmittance (%)	22.1 ± 0.6	21.6 ± 0.7	21.2 ± 0.8	20.8 ± 0.9	−4.6 ± 0.8
crack risk probability (%)	<0.1	0.8 ± 0.3	2.4 ± 0.6	5.2 ± 1.1	+5.1 ± 1.1
interfacial debond area (%)	0.3 ± 0.1	0.8 ± 0.2	1.4 ± 0.3	2.1 ± 0.4	+1.8 ± 0.4
maintenance trigger point	N/A	none	none	26.4 ± 1.8	N/A
RMSE (modulus prediction)	N/A	0.12 GPa	0.15 GPa	0.18 GPa	N/A
RMSE (transmittance prediction)	N/A	0.14%	0.18%	0.21%	N/A
RMSE (temperature prediction)	N/A	0.8 °C	0.9 °C	1.1 °C	N/A

https://doi.org/10.1021/acsomega.5c10292
ACS Omega 2026, 11, 8011−8023

costs by 15−20% with industrial production volumes, making it increasingly competitive with conventional systems requiring multiple separate components to achieve similar functionality.

4.2. Energy Efficiency and Sustainability

Our comparative lifecycle analysis shows that the GFRP-TRPC system consumes 330 W (285 W for the LED system and 45 W for the SHM system) but eliminates the need for separate façade lighting (typically 450−600 W for an equivalent area) and dedicated SHM systems (80−120 W). The integrated approach yields a net energy savings of approximately 200−390 W per panel. Additionally, the daylighting benefit reduces interior artificial lighting needs by 22−28% during daylight hours, providing further energy savings of 120−180 kWh annually per panel.

We are currently developing solar-powered LED backlighting that could reduce operational energy by 65−80%. A full lifecycle energy analysis shows that the GFRP-TRPC system achieves energy payback within 4.3 years compared to conventional systems when accounting for embodied and operational energy.

Our lifecycle assessment shows that the GFRP-TRPC system has an embodied carbon footprint of 78.3 kg of CO_2eq/m^2 compared to 102.7 kg of CO_2eq/m^2 for conventional cement-based translucent concrete and 124.5 kg of CO_2eq/m^2 for aluminum curtain wall systems with comparable functionality. The 30% recycled glass content contributes to a 12.4% reduction in embodied carbon compared to virgin materials. The enhanced daylighting performance reduces artificial lighting energy by 22−28%, resulting in operational carbon savings of approximately 18.7 kg of CO_2eq/m^2/year. This yields a carbon payback period of 3.2 years compared to conventional systems. End-of-life scenarios were also modeled, material recovery through specialized composite recycling processes. This recovery rate significantly improves the overall lifecycle sustainability profile of the GFRP-TRPC system compared to conventional building envelope solutions.

The use of recycled glass aggregate not only reduces embodied carbon but also diverts material from landfills, contributing to circular economy principles. Each square meter of the GFRP-TRPC panel incorporates approximately 14.2 kg of recycled glass, equivalent to approximately 35 recycled bottles. The potential for further increasing recycled content is being investigated, with preliminary tests showing promising results for up to 45% recycled glass content without significant degradation of mechanical or optical properties.

4.3. Comparison with Commercial Alternatives

Our analysis shows the following advantages compared to commercial alternatives: (1) Versus ETFE cushions with integrated sensors: 280% higher structural capacity and 40% lower thermal transmittance but 35% higher initial cost. The GFRP-TRPC system eliminates the need for pneumatic pressure maintenance systems required by ETFE solutions, reducing long-term operational costs and failure risks. (2) Versus translucent concrete with separate LED panels: 26% weight reduction, 18% higher optical transmittance, 40% lower installation complexity, and 12% higher cost amortized over a 30-year lifespan. The integrated approach eliminates the need for separate mounting systems and reduces the overall wall thickness by 35−45%. (3) Versus electrochromic glazing with separate monitoring: 85% higher impact resistance, comparable energy performance, but 28% higher initial cost. The GFRP-TRPC system offers superior durability in high-traffic areas and eliminates the risk of complete functionality loss due to power failure that affects electrochromic systems.

The GFRP-TRPC system represents a premium solution with a higher initial cost but offers integrated functionality that eliminates separate systems and provides superior long-term performance. The 30-year lifecycle cost analysis shows that the total cost of ownership is 8−15% lower than equivalent conventional systems when accounting for installation, maintenance, and replacement costs.

4.4. Human-Building Interaction and Occupant Experience

The AI-driven light narration system demonstrates a new paradigm for human-building interactions that goes beyond conventional responsive architecture. By incorporating pose estimation and generative adversarial networks, the system creates dynamic light patterns that respond to human movement in aesthetically meaningful ways rather than through simplistic trigger-response mechanisms.

The observed 340% increase in visitor dwell time during exhibition testing indicates the powerful engagement potential of such systems. More importantly, the preliminary studies in office and healthcare environments suggest broader applications beyond artistic installations. The ability to calibrate interaction sensitivity and visual impact for different functional contexts enables the system to enhance the occupant experience without causing distraction or fatigue.

The integration of structural health monitoring with an aesthetic display creates a novel safety communication channel that leverages human visual perception. Traditional SHM systems typically alert only facility managers through technical dashboards, while our approach enables direct communication with building occupants through intuitive visual cues. The human factor testing confirmed that this approach is effective across diverse demographics and does not require specialized training to interpret.

For global deployment across diverse cultural contexts, we are developing culturally adaptive interaction modes that can be calibrated to local aesthetic preferences and social norms regarding personal space and public interaction. Preliminary testing with focus groups from five different cultural backgrounds (n = 87 participants) showed that cultural calibration improved acceptance rates by 27−35% compared to a one-size-fits-all interaction design.

4.5. Digital Twin Accuracy and Long-Term Reliability

The digital twin framework demonstrates strong predictive capability for both short-term responses and long-term degradation. The validation against accelerated aging tests and the 14-month real-world installation provides confidence in the 30-year service life predictions. However, several limitations and opportunities for improvement should be acknowledged. First, the prediction uncertainty increases with time horizon, following approximately a $t^{0.58}$ relationship. This underscores the importance of periodic recalibration based on the actual performance data. Our analysis suggests that 24-month recalibration intervals are optimal for maintaining the prediction accuracy within ± 10% for 5+ year forecasts. Second, the current digital twin focuses primarily on mechanical and optical degradation but could be expanded to include additional performance aspects such as thermal bridging, acoustic properties, and fire resistance. These extensions would provide a more comprehensive lifecycle management tool for building owners and facility managers. Third, the integration between the digital twin and building

https://doi.org/10.1021/acsomega.5c10292
ACS Omega 2026, 11, 8011−8023

management systems could be enhanced to enable automated adjustment of maintenance schedules based on actual performance rather than predetermined intervals. This would optimize maintenance resources and minimize disruption to building operations. The long-term reliability of embedded FBG sensors is critical for the continued accuracy of the digital twin. Our accelerated aging tests project sensor functional lifetime >25 years before signal-to-noise ratio falls below the 10:1 threshold required for reliable crack detection. However, we recommend a redundant sensor layout with 30% additional FBGs to ensure system reliability even with individual sensor failures. Our maintenance protocol includes quarterly recalibration using reference load cases and annual diagnostic testing to identify sensor degradation.

4.6. Future Research Directions

Several promising research directions emerge from this work: (1) Adaptive Material Formulations: Development of climate-specific GFRP-TRPC formulations with tailored additives is needed for extreme environments. For global deployment, we recommend climate-specific formulation adjustments: increased UV stabilizers for equatorial regions, modified toughening agents for arctic applications, and enhanced hydrophobic treatments for marine environments. (2) Advanced AI Integration: Exploration of reinforcement learning approaches that allow the system to optimize interaction patterns based on occupant feedback and engagement metrics over time creates a truly adaptive building skin that evolves with use. (3) Expanded Sensing Capabilities: Integration of additional sensing modalities such as air quality monitoring, occupancy detection, and thermal imaging creates a more comprehensive environmental monitoring system within the building envelope. (4) Bioinspired Adaptive Responses: Development of biomimetic response patterns that emulate natural systems such as plant phototropism or animal chromatophores creates building skins that respond to environmental conditions in biologically inspired ways. (5) Decentralized Digital Twins: Investigation of federated digital twin networks allows multiple building components to share data and coordinate responses, creating emergent intelligence at the building system level rather than isolated component optimization.

5. CONCLUSIONS

This study demonstrates a comprehensive approach to creating multifunctional building skins that integrate structural performance, optical functionality, and intelligent interaction. The GFRP-reinforced translucent polymer-concrete (GFRP-TRPC) panel successfully combines load-bearing capacity, daylighting, and AI-driven interaction in a 30-mm-thin composite element. Key innovations and findings include (1) a scalable manufacturing process using vacuum-assisted resin infusion that achieves exceptional microstructural quality with void fractions below 1.2%, enabling both high mechanical performance (65 MPa compressive strength and 48.7 MPa flexural strength) and optical functionality (22% transmittance).

Integration of bidirectional GFRP reinforcement with PMMA optical fibers in a polymer-concrete matrix created a lightweight (1.78 g/cm^3) composite with 26% weight reduction compared to conventional translucent concrete while maintaining superior mechanical properties.

Development of an AI-driven light narration system using Vision Transformer−GAN architecture that converts human movement into dynamic light patterns created an interactive "living fresco" that increased visitor engagement by 340% in exhibition settings and showed promising results in functional building contexts.

Implementation of a distributed FBG sensor network with edge-deployed TinyML for real-time crack detection (96.7% accuracy for 0.1 mm cracks) and safety communication through visual alerts creates a self-sensing building skin that communicates structural health to occupants.

Creation of a calibrated digital twin framework that predicts only 7.8% stiffness loss and <5% optical attenuation after 30 years of service, with maintenance alerts triggered six months in advance of critical thresholds, enabling proactive lifecycle management.

The GFRP-TRPC system demonstrates that advanced polymer composites can serve as platforms for intelligent building skins that merge structural integrity, energy-efficient daylighting, and human-centric digital interaction. The approach represents a significant step toward cyberphysical architectural elements that actively participate in building performance, occupant experience, and lifecycle management.

Future work will focus on climate-specific material formulations, expanded sensing capabilities, and more sophisticated AI integration to further enhance the adaptive capabilities of the system.[30,31] The principles and technologies demonstrated in this study have broad applications beyond building façades, including transportation infrastructure, public art installations, and healthcare environments, where the combination of structural performance, aesthetic quality, and intelligent monitoring provides significant advantages over conventional materials.

AUTHOR INFORMATION

Corresponding Author

Feng Han − *School of Smart City, Shanghai Construction Management Vocational College, Shanghai 201702, P. R. China;* orcid.org/0009-0004-6842-099X; Email: hanfeng@fudan.edu.cn

Authors

Shu Wu − *Sino-German College, University of Shanghai for Science and Technology, Shanghai 200093, P. R. China;* orcid.org/0009-0006-6971-4118

Qi Chen − *Sino-German College, University of Shanghai for Science and Technology, Shanghai 200093, P. R. China*

Chao Hu − *School of Civil and Environmental Engineering, Nanyang Technological University, Singapore 639798, Singapore;* orcid.org/0009-0005-4597-7995

Yingjian An − *School of Smart City, Shanghai Construction Management Vocational College, Shanghai 201702, P. R. China;* orcid.org/0000-0002-8945-1451

Complete contact information is available at: https://pubs.acs.org/10.1021/acsomega.5c10292

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This research was supported by the National Natural Science Foundation of China (Grant No. 52078301), the Shanghai

https://doi.org/10.1021/acsomega.5c10292
ACS Omega 2026, 11, 8011−8023

Municipal Science and Technology Commission (Grant No. 20ZR1460500), the Sino-German Center for Research Promotion (Grant No. GZ1480), National Social Science Foundation (Grant No. 24BGL078), the Fund of Shanghai Education Science Research Project (Grant No. C2024196); The Ministry of Education's Industry University Cooperation Collaborative Education Project (Grant No. 231101913174352); The Ministry of Education's Supply-Demand Docking Employment and Education Project (Grant No. 2023123031821); Funds of University of Shanghai for Science and Technology (Grant Nos. YLKC202424443, CFTD2025ZD16, SH2025268, SH2025265, and XJ2025564), 2025 Special Project for Women's Civilization Construction, 2025 Practical Innovation Project for Academic Atmosphere Construction; 2025 "Social Innovation and Education Integration" Community Education Innovation Project (Grant No. SCRJ20250501005); the Humanities and Social Sciences Cultivation Fund Project of USST (Grant No. 24SKPY04); The Talent Development Model for High-Skilled Professionals in China Vocational Education Society 2025 Annual Planning Research Project (Grant No. ZJS2025ZD36), and The China Youth & Children Research Association "Climb Plan" (Grant No. G2025-0901-005).

REFERENCES

(1) Loonen, R. C. G. M.; Trčka, M.; Cóstola, D.; Hensen, J. L. M. Climate adaptive building shells: State-of-the-art and future challenges. *Renewable Sustainable Energy Rev.* **2013**, *25*, 483−493.
(2) Romano, R.; Aelenei, L.; Aelenei, D.; Mazzucchelli, E. S. What is an adaptive façade? A comprehensive review of the concept and its performance assessment. *Energy Build.* **2024**, *227*, No. 110396.
(3) Kolarevic, B.; Parlac, V. *Building Dynamics: Exploring Architecture of Change*; Routledge, 2023.
(4) He, J.; Zhou, Z.; Ou, J.; Huang, M. Numerical and experimental study on mechanical properties of transparent concrete. *Constr. Build. Mater.* **2022**, *179*, 10−17.
(5) Altlomate, A.; Alatshan, F.; Mashiri, F.; Jadan, M. Experimental study on light-transmitting concrete (LTC) produced with plastic optical fiber and glass optical fiber. *Case Stud. Constr. Mater.* **2023**, *14*, No. e00492.
(6) Saleem, M. A.; Azam, A.; Kazmi, S. M. S.; Munir, M. J. Development, mechanical properties and numerical modeling of transparent concrete: A state-of-the-art review. *Constr. Build. Mater.* **2022**, *314*, No. 125596.
(7) Shafigh, P.; Mahmud, H. B.; Jumaat, M. Z.; Zargar, M. Lightweight concrete as a sustainable material: A review. *Constr. Build. Mater.* **2023**, *169*, 37−49.
(8) Majumdar, A.; Banerjee, S.; Sanyal, D.; Dutta, A. Degradation of optical fibers in alkaline environment: Implications for smart concrete applications. *Cem. Concr. Compos.* **2022**, *117*, No. 103867.
(9) Luna-Navarro, A.; Meizoso, M.; DeBleecker, H.; Verhoeven, Y.; Overend, M. Façade intelligence: A comparative analysis of smart, intelligent and cognitive façade systems. *Energy Build.* **2023**, *229*, No. 110475.
(10) Guo, M. Z.; Chen, Z.; Liu, T. C.; Poon, C. S.; Xiao, J. How to predict service life of innovative construction materials? A critical review. *Constr. Build. Mater.* **2024**, *345*, No. 128764.
(11) Bank, L. C.; Gentry, T. R.; Thompson, B. P.; Russell, J. S. A model specification for FRP composites for civil engineering structures. *Constr. Build. Mater.* **2023**, *185*, 884−890.
(12) Correia, J. R.; Bai, Y.; Keller, T. A review of the fire behaviour of pultruded GFRP structural profiles for civil engineering applications. *Compos. Struct.* **2015**, *127*, 267−287.
(13) Minakuchi, S.; Takeda, N. Recent advances in optical fiber sensing for composite structures. *Compos. Struct.* **2023**, *171*, 85−100.
(14) Ghosh, S.; Chakraborty, S. Recent trends in smart composite materials with embedded sensors and actuators. *J. Intell. Mater. Syst. Struct.* **2022**, *33* (2), 267−298.
(15) Majumder, M.; Gangopadhyay, T. K.; Chakraborty, A. K.; Dasgupta, K.; Bhattacharya, D. K. Fibre Bragg gratings in structural health monitoring—Present status and applications. *Sens. Actuators, A* **2008**, *147* (1), 150−164.
(16) Di Sante, R. Fibre Optic Sensors for Structural Health Monitoring of Aircraft Composite Structures: Recent Advances and Applications. *Sensors* **2015**, *15* (8), 18666−18713.
(17) Worden, K.; Cross, E. J.; Gardner, P. Structural health monitoring as a big data problem. *Mech. Syst. Signal Process.* **2024**, *163*, No. 108162.
(18) Kretzer, M.; Hovestadt, L. ALIVE: Advancements in adaptive architecture. *Comput.-Aided Des.* **2023**, *121*, No. 102807.
(19) Addington, M.; Schodek, D. L. *Smart Materials and New Technologies: For the Architecture and Design Professions*; Routledge, 2022.
(20) Tao, F.; Zhang, H.; Liu, A.; Nee, A. Y. C. Digital Twin in Industry: State-of-the-Art. *IEEE Trans. Ind. Inf.* **2019**, *15* (4), 2405−2415.
(21) Lu, Q.; Parlikad, A. K.; Woodall, P.; Don Ranasinghe, G.; Heaton, J. Developing a dynamic digital twin at a building level: Using IoT and machine learning technologies to predict building energy consumption. *Adv. Eng. Inf.* **2024**, *43*, No. 101055.
(22) Khajavi, S. H.; Motlagh, N. H.; Jaribion, A.; Werner, L. C.; Holmström, J. Digital Twin: Vision, benefits, boundaries, and creation method. *J. Ind. Inf. Integr.* **2023**, *26*, No. 100262.
(23) Seyedzadeh, S.; Rahimian, F. P.; Glesk, I.; Roper, M. Machine learning for estimation of building energy consumption and performance: a review. *Visualization in Eng.* **2018**, *6* (1), No. 5.
(24) Park, J. Y.; Ouf, M. M.; Gunay, B.; Peng, Y.; O'Brien, W.; Kjærgaard, M. B.; Nagy, Z. A critical review of field implementations of occupant-centric building controls. *Build. Environ.* **2024**, *165*, No. 106369.
(25) Zou, H.; Zhou, Y.; Yang, J.; Spanos, C. J. Towards occupant-oriented smart buildings: Recent advances in occupancy sensing and analytics. *Energy Build.* **2023**, *216*, No. 109942.
(26) Cudzik, J.; Radziszewski, K. Generative adversarial networks in architectural design: A comprehensive review. *Autom. Constr.* **2024**, *127*, No. 103649.
(27) Dey, A.; Choudhury, M. D. Recent developments in translucent concrete: Materials, methods, and applications. *Constr. Build. Mater.* **2023**, *287*, No. 122820.
(28) Mathur, P. *Smart Buildings: Advanced Materials and Nanotechnology to Improve Energy-Efficiency and Environmental Performance*; Woodhead Publishing, 2022.
(29) Bribián, I. Z.; Capilla, A. V.; Usón, A. A. Life cycle assessment of building materials: Comparative analysis of energy and environmental impacts and evaluation of the eco-efficiency improvement potential. *Build. Environ.* **2011**, *46* (5), 1133−1140.
(30) Li, R. D.; Feng, S. M. Academic librarians in research competitions: A network analysis of resource provision and collaborative dynamics. *J. Acad. Libr.* **2025**, *51*, No. 103100.
(31) Li, R. D.; Zhang, J. A holistic evaluation framework for Chinese graded reading system. *J. Data Inf. Sci.* **2025**, *10* (4), 197−218.

https://doi.org/10.1021/acsomega.5c10292
ACS Omega 2026, 11, 8011−8023